

Sami Aljabery

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Education

M.Eng. in Electrical and Computer Engineering	Jun. 2026
Oregon State University Corvallis, OR	GPA: 3.62
• Relevant Coursework: HPC Architecture, Operating Systems II, Modern Memory Systems, VLSI System Design, Analog CMOS Integrated Circuits, Parallel Programming, Advanced DL and HPC, Principles of Computer System Design, Machine Learning	
B.S. in Electrical and Computer Engineering Minor in Computer Science	Dec. 2024
Oregon State University Corvallis, OR	GPA: 3.51

Experience

XCP Parallel Computing Research Fellow	Jun. 2026 – Present
Los Alamos National Laboratory Los Alamos, NM	
• Accelerate the xRAGE Eulerian multiphysics production code for Inertial Confinement Fusion (ICF) capsule simulations on LANL's Rocinante HPC cluster through targeted performance profiling with OpenMP and the Kokkos portability layer.	
• Diagnose write-contention bottlenecks in C++ production code via Perfetto traces, comparing Kokkos atomic operations against ScatterView duplication-reduction to minimize thread synchronization overhead in parallel loops.	
Graduate Teaching Assistant	Jan. 2025 – Jun. 2026
Oregon State University Corvallis, OR	
• Managed system verification for 16+ ECE capstone teams, testing hardware/software blocks against design specifications using oscilloscopes, waveform generators, power supplies, electronic loads, and testbenches.	
• Conducted weekly Computer Organization and Assembly recitations on AVR assembly and ATmega32U4 microcontroller projects, introducing concepts such as pulse width modulation and interrupts.	
Digital Design Engineering Intern	Jun. 2025 – Sep. 2025
onsemi Corvallis, OR	
• Validated the Verilog RTL of AR0827 image sensor's non-deterministic random bit generator with Python testbenches, verifying entropy quality and oscillator stability to confirm readiness for PVT characterization.	
• Designed and routed a custom PCB to synchronize image sensors via Raspberry Pi and logic level shifting for high-speed shutter validation.	
EECS Undergraduate Learning Assistant	Mar. 2022 – Dec. 2024
Oregon State University Corvallis, OR	
• Facilitated recitations and labs for 60+ students in Electrical Fundamentals (I/II) and Digital Logic Design.	
• Mentored Digital Logic Design students using SystemVerilog and ModelSim simulation workflows, verifying FPGA implementations including VGA drivers on DE10-Lite FPGA boards through Quartus Prime.	

Leadership

Embedded Systems Design Club President	Aug. 2024 – Jun. 2026
• Founded the Embedded Systems Design Club (ESDC) at Oregon State University, hosting 20+ meetings, workshops, and projects emphasizing hardware knowledge, communication protocols, and debugging.	

Projects

RAID-5 Distributed Storage System	May. 2026 – Jun. 2026
• Architected a fault-tolerant RAID-5 distributed file system in Python using checksum validation and byte-by-byte XOR parity generation to detect data corruption, reconstructing lost blocks during single server fail-stop events.	
• Profiled system performance against a baseline single server case using two different block sizes and file sizes, demonstrating a 76% reduction in average server load for 6-server configuration.	
ATmega32U4 Real-Time Sensor System	Nov. 2024 – Dec. 2024
• Engineered a real-time monitoring system in C on a custom ATmega 32U4 board, writing custom I2C and LCD drivers to interface with TC74 thermal sensors, an ultrasonic sensor, and an onboard LCD.	
• Established an interrupt-driven timing system by configuring two hardware timers, one in CTC mode to generate a real-time clock and another in input capture to measure ultrasonic sensor echo pulses for distance calculations.	
DE10-Nano FPGA Handheld Console	Sep. 2023 – Jun. 2024
• Directed a 4-person team delivering a proof-of-concept FPGA handheld utilizing open-source MiSTer hardware emulation.	
• Managed system integration to ensure portability, incorporating custom PCB power system delivering 2+ hours of battery life.	
• Awarded the Engineering Excellence Award in recognition of prototyping and technical execution.	

Skills

Languages: Python, C, C++, Verilog, SystemVerilog, MATLAB, AVR Assembly
Hardware & Protocols: FPGAs, Microcontrollers, PCB Design, SPI, I2C, UART
Software & Computing: Linux, PyTorch, MPI, OpenMP, CUDA, OpenCL, Kokkos
Tools: Cadence Virtuoso, Quartus Prime, Xilinx Vivado, Autodesk Fusion 360, KiCAD, QEMU, GDB, Git, Perfetto